Taicheng Guo

Updated October 31, 2026

Email: taicheng@amazon.com **Homepage** **Google Scholar** **Github**

RESEARCH INTERESTS

My long-term research & engineering goal is to build systems for AGI. My research centers on **Agents** and **Systems**, and how they co-evolve on the path to superintelligence, especially *Effective Long-Horizon **Agentic Training*** for *AI-System / Complex-Task Automation*, ***Agent-System Co-Design*** *(Inference Optimization)*, and *Language Model **Post-Training***. I work as a research engineer who is driven to build novel & fundamental systems from scratch.

WORK&INTERN EXPERIENCE

Amazon (Applied Scientist II) Oct. 2026 – present
Team: Amazon Stores Foundation AI.
Topic: Training and Building Intelligent Conversational Agents For Alexa for Shopping (Rufus).

ND-IBM Research Collaboration (Research Assistant) Jun. 2026 – present
Topic: Training Long-Horizon Agents with Self-Improving Verifiers for open-world problems.

Amazon (Applied Scientist Intern) Jan. 2025 – Jul. 2025
Topic: Training Long-Horizon Coding Agents. **[ACL 2026 Main]**

Amazon (Applied Scientist Intern) Jun. 2024 – Sep. 2024
Topic: Building Long-Horizon Agents within Database for Personalization. **[KDD@RelSciFM]**

MBZUAI (Research Intern) Aug. 2022 – Oct. 2022
Topic: Training Conditional-Based Diffusion Model. **[ICASSP 2024]**

KAUST (Visiting Student) Dec. 2020 – Aug. 2021
Topic: Training Language Models for Cross-Lingual News Recommendation. **[WWW 2023 Oral]**

Tuyoo Games (Startup), AI Lab (Senior Research Engineer) Aug. 2018 – Oct. 2020
Topic: Led a 4-person team on Game-AI intelligence, building zero-sum game agents via Monte Carlo Tree Search (MCTS), Minimax, etc. **[Employee of the Year Award (Top 2%)]**

Qunar (Strategy Product Manager) Topic: Personalization Jul. 2017 – May. 2018

EDUCATION

University of Notre Dame (Jan. 2023 – Sep. 2026) US
Ph.D. in Computer Science and Engineering Advisor: Prof. Xiangliang Zhang. *GPA: 4.0/4.0*
Thesis: From Answering to Autonomous Research: Language Models for Research-Level Tasks
Committee: Profs. Xiangliang Zhang, Nitesh V. Chawla, Meng Jiang, Wei Wang (UCLA).

King Abdullah University of Science and Technology (KAUST) Saudi Arabia
M.S. in Computer Science Advisor: Prof. Xiangliang Zhang. *GPA: 3.89/4.0*

Xidian University Bachelor in Software Engineering. *GPA: 3.5/4.0*

ACADEMIC COMPETITIONS

KDD Cup Large-Scale RecSys Competition. (**2nd / 1895 teams, $4,000**) [Link]
NeurIPS Black-Box AutoML Optimization Challenge. (**2nd / 200+ teams , W.S. track**) [Link]
Kaggle Text Sentiment Analysis. (**5th / 80+ teams**) & **Kaggle Expert** [Link]
IEEE Computer Society Data Science challenge. (**1st place, $1000**) [Link]

SELECTED PUBLICATIONS

▹ Citations: **4009** (80%+ from first-author papers) **Full Paper List**

Long-Horizon Agentic Training & Automated Research

[1] **AutoLLMResearch: Training Research Agents for Automating LLM Experiment Configuration – Learning from Cheap, Optimizing Expensive**
Taicheng Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In Submission

[2] **MTSQL-R1: Towards Long-Horizon Multi-Turn Text-to-SQL via Agentic Training**

Taicheng Guo, Hai Wang, ChaoChun Liu, Mohsen Golalikhani, Xin Chen, Xiangliang Zhang, Chandan K. Reddy
In ACL 2026 Main, Acceptance rate: 19%, 40+ Github Stars

Agent-System Co-Design & Multi-Agent Systems

[3] **ASAP: Agent-System Co-Design for WallTime-Centered Auto HPO Research for ML Experiments**
Taicheng Guo, Haomin Zhuang, Kehan Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In Submission

[4] **Large Language Model based Multi-Agents: A Survey of Progress and Challenges**
Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In IJCAI 2024, Acceptance rate: 20%, Top-1 Most Influential Paper, 1k+ Github Stars

[5] **Data Interpreter: An LLM Agent For Data Science**
Sirui Hong, ... **Taicheng Guo** as a contributor to Graph-based Agent Planning.
In ACL Findings 2025, 68k+ Github Stars

[6] **Knowledge Graph Enhanced Language Agents for Recommendation**
Taicheng Guo, Chaochun Liu, Hai Wang, Varun Mannam, Fang Wang, Xin Chen, Xiangliang Zhang, Chandan K. Reddy
In KDD@RelSciFM Workshop Oral

Language Model Post-Training, Reasoning and Applications

[7] **ReactionTeam: Teaming Experts for Divergent Thinking Beyond Typical Reaction Patterns.**
Taicheng Guo, Changsheng Ma, Xiuying Chen, Bozhao Nan, Kehan Guo, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In IEEE Conference on BigData 2025 Oral, Acceptance rate: 18.7%

[8] **What can Large Language Models do in chemistry? A comprehensive benchmark on eight tasks**
Taicheng Guo, Kehan Guo, Bozhao Nan, Zhenwen Liang, Zhichun Guo, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang
In NeurIPS 2023, Acceptance rate: 32.7%, Top 2% Influential Paper, 150+ Github Stars

[9] **Few-shot News Recommendation via Cross-lingual Transfer**
Taicheng Guo, Lu Yu, Basem Shihada, Xiangliang Zhang
In WWW 2023 Oral, Acceptance rate: 19.2%

[10] **Domain-Adaptive Reasoning model for Injury Classification from Incident Descriptions**
Taicheng Guo, Hai Wang, Shucheng (Sue) Zhuang, Mohsen Golalikhani, Chaochun Liu, Wei Wang, Xin Chen
In AMLC 2025 RL workshop, Cross-team collaboration work

PROFESSIONAL SERVICE

Reviewer: NeurIPS, ICLR, ACL, EMNLP, WWW, KDD, IJCAI, TPAMI, Nature Comm., TKDE, CSUR
Teaching Assistant: CSE 40625 Machine Learning, University of Notre Dame, 2023